Multitask Factor Analysis with Application to Noise Robust Radar HRRP Target Recognition

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***Abstract*— A factor analysis model based on multitask learning (MTL) is developed to characterize the FFT-magnitude feature of complex high-resolution range profile (HRRP), motivated by the problem of radar automatic target recognition (RATR). The MTL mechanism makes it possible to appropriately share the information among samples from different target-aspects and learn the aspect-dependent parameters collectively, thus offering the potential to improve the overall recognition performance with small training data size. In addition, since the noise level of a test sample is usually different from those of the training samples in the real application, another contribution is that the proposed framework can update the noise level parameter in the FA model to adaptively match that of the received test sample. Efficient inference is performed via variational Bayes (VB) for the proposed hierarchical Bayesian model, and encouraging results are reported on the measured HRRP dataset with small training data size and under the test condition of low signal-to-noise ratio (SNR).**

I. Introduction

Radar high-resolution range profile (HRRP) has received intensive attention from the radar automatic target recognition (RATR) community [3]–[6], [8]–[11], [14]. According to the definition of HRRP given in these literatures, an HRRP is a *real* vector, which is composed of the amplitude of the coherent summations of the complex returns from target scatterers in each range cell. For radar HRRP recognition, as discussed in the literatures [5], [6], some pre-processing techniques are needed to deal with the target-aspect, time-shift and amplitude-scale sensitivity problems. Feature extraction from HRRP is a key step in the recognition system. Paper [9] investigates the bispectra feature extracted from real HRRP data, and [5] further compares the recognition performance of various high-order spectra features also extracted from real HRRP data. In [10], [14], some superresolution algorithms are employed to extract the location information of predominant scatterers from real HRRP data as feature. Nevertheless, some information contained in the HRRP samples is inevitably lost during the dimensional reduction procedure. As demonstrated in [7], compared with the real HRRP data, complex HRRP data reserve some phase information useful for target discrimination. Nevertheless, there has been little work done in the field of RATR using complex HRRPs. We believe that the reason is complex HRRPs' strong sensitivity to target position variation, which is referred to as the initial-phase sensitivity. Therefore, the initial-phase invariant feature extraction is a precondition for statistical modeling for complex HRRP data.

Some parametric statistical models are developed to describe the class-conditional likelihood of real HRRP data, *e.g.,* Gaussian model [11], Gamma model [3], the two-distribution compounded model comprising Gamma distribution and Gaussian mixture distribution [6] and factor analysis (FA) model [4]. In addition, the hidden Markov model (HMM) has been utilized by [10], [14] for radar HRRP sequence recognition. Nevertheless, there are two limitations of these statistical models. Firstly, they usually require substantial training data, especially for the complicated statistical models with many unknown parameters, such as FA model in [4] and HMM model in [10], [14]. However, due to the target-aspect sensitivity of HRRP data, it is hard to collect enough training data for each target-aspect of each target. Rather than building models for each data subset associated with different target-aspect individually, it is desirable to appropriately share the information among these related data, thus offering the potential to improve overall recognition performance. If the modeling of one data subset is termed one learning task, the learning of models for all tasks jointly is referred to as multitask learning (MTL) [2]. Our recent work in [8] proposed an HMM-based MTL algorithm for real HRRP samples and inspiring recognition performance is achieved only with small training data size. Secondly, all the existing statical models for HRRP data assume that the training data and test data are obtained under the similar measurement circumstance. In real application, the training data are usually collected under the condition of high signal-to-noise ratio (SNR) via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to the severe measurement conditions, such as the large distance between the non-cooperative targets and radar. Thus the statistical model for RATR should be modified to match the noise level of the received test sample in the classification stage to obtain the noise-robust recognition performance.

The research reported here seeks a noise-robust statistical recognition method based on MTL for complex HRRP. In this paper, we develop a *noise-robust factor analysis model based on multitask learning* (noise-robust MTL-FA) for the FFT-magnitude feature extracted from complex HRRP data, where the initial-phase and time-shift invariant feature makes it possible to share the same factor loading matrix across

This work was partially supported by the National Science Foundation of China (NO.60901067 and NO.61001212).

978-1-4673-0658-4/12/$31.00 ©2012 IEEE

different target-aspects of each target, but the latent factor variables are target-aspect dependent. Another contribution is that the proposed framework can update the noise level parameter in the FA model in the classification stage to adaptively match that of the received test sample. In this way, the recognition performance under the test condition of low SNR can be improved. The form of the proposed hierarchical model allows efficient variational Bayesian (VB) inference, of interest for large-scale problems. The RATR experiments based on measured HRRP dataset show that the proposed model not only has good recognition performance with small training data size but also is more robust to noise than other existing statistical models.

II. Bayesian Model Construction

A. Feature extraction from complex HRRP data

Suppose there are radial displacement and little uniform rotation for a target. The nth complex returned echo from the lth range cell ($l = 1, 2, \cdots, L$) is

$$\begin{aligned} x_l(n) &= \sum_{i=1}^{V_l} \sigma_{li} \exp\{-j[\frac{4\pi}{\lambda}(R(n) + \Delta r_{li}(n)) - \theta_{li0}]\} \\ &= e^{-j\frac{4\pi}{\lambda}R(n)} \sum_{i=1}^{V_l} \sigma_{li} \exp\{-j[\frac{4\pi}{\lambda}\Delta r_{li}(n) - \theta_{li0}]\} \\ &= e^{j\theta(n)} \sum_{i=1}^{V_l} \sigma_{li} e^{j\phi_{li}(n)} \end{aligned} \tag{1}$$

where λ denotes the radar wavelength, V_l denotes the number of target scatterers in the lth range cell, σ_{li} and θ_{li0} represent the strength and the initial phase of the ith scatterer in the lth range cell, $R(n)$ denotes the radial distance between the target reference center in the nth returned echo and the radar, and $\Delta r_{li}(n)$ represents the radial displacement of the ith scatterer in the lth range cell in the nth returned echo. In (1), $\theta(n) = -\frac{4\pi}{\lambda}R(n)$ denotes the initial phase of the nth returned echo, which is strongly sensitive to the target position variation. For example, for carrier frequency of 5520 MHz (approximately, the wavelength of 5 cm), if the radial displacement of a target is 2.5 cm, the initial phase difference will be 2π. In real applications, the measurement precision of the target distance usually cannot satisfy the requirement of the initial phase compensation, therefore, the measured complex HRRP data have the initial-phase sensitivity.

Moreover, since an HRRP sample is only a part of received radar echo extracted by a range window, in which target signal is included, the position of the target signal in an HRRP sample can vary with the measurement. This is the time-shift sensitivity. Considering these two sensitivity issues of the complex HRRP data, *i.e.*, the initial-phase sensitivity and time-shift sensitivity, we prefer to extracting a feature which is independent of the complex HRRP's initial-phase and time-shift, and then characterizing such feature via a MTL statistical model in this paper.

References [12], [13] propose to use FFT-magnitude feature for radar HRRP target recognition. The frequency spectrum $\boldsymbol{\mathcal{F}}(n)$ of a complex HRRP $\boldsymbol{x}(n)$ is

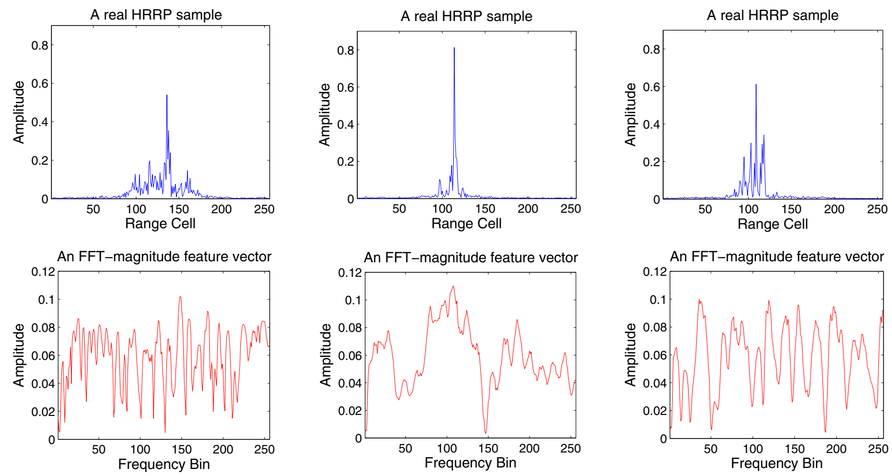
$$\begin{aligned} \boldsymbol{\mathcal{F}}(n) &= \mathrm{FFT}(\boldsymbol{x}(n)) \\ &= e^{j\theta(n)}[\mathcal{F}_1(n), \mathcal{F}_2(n), \cdots, \mathcal{F}_L(n)]^{\mathrm{T}} \\ &= e^{j\theta(n)}[\sum_{l'=1}^{L} x_{l'}(n)e^{\frac{-j2\pi\times 1}{L}}, \sum_{l'=1}^{L} x_{l'}(n)e^{\frac{-j2\pi\times 2}{L}}, \\ &\quad \cdots, \sum_{l'=1}^{L} x_{l'}(n)e^{\frac{-j2\pi\times L}{L}}]^{\mathrm{T}} \end{aligned} \tag{2}$$

where the function $\mathrm{FFT}(\cdot)$ denotes the fast Fourier transform (FFT). From (2) we can see the frequency spectrum $\boldsymbol{\mathcal{F}}(n)$ still suffers from the time-shift sensitivity and initial-phase sensitivity. Nevertheless, according the property of FFT, the time-shift of $\boldsymbol{x}(n)$ can only bring a new phase term to its frequency spectrum. Therefore, same as papers [12], [13], here we only use the amplitude of frequency spectrum $\boldsymbol{y}(n) = |\boldsymbol{\mathcal{F}}(n)|$, *i.e.*, FFT-magnitude, as the feature vector for the following statistical modeling. Obviously, the FFT-magnitude feature is independent of the complex HRRP's time-shift and initial-phase, which is referred to as the time-shift and initial-phase invariant property.

As discussed in [12], [13], there are many advantages of working in the frequency domain. Firstly, it is found that the frequency domain target signatures are distinct for different aircraft types and at different aspects. As a result, target identification in the frequency domain is just as viable as using conventional real HRRPs in the range domain. Secondly, since FFT is an information preserving procedure, there is no information lost in the transformation process. Thirdly, the FFT-magnitude feature of complex HRRP has the time-shift and initial-phase invariant property. Finally, for the stepped frequency waveform (SFWF) radar system, it is much easier to get the frequency spectrum of moving targets. Usually, to obtain the HRRP sample from the SFWF, the radial motions along the radar line of sight (RLOS), the velocity and acceleration of the target have to be compensated; otherwise, broadening of the HRRP envelope will occur. Figure 1 presents the real HRRPs and corresponding FFT-magnitude feature vectors of 3 airplane targets at the same target-aspect. It is evident from Figure 1 that the echo amplitude of the frequency spectra are just as distinct as the corresponding range profiles.

B. Statistical modeling

For the motivating RATR problem of interest here, one typically has access to multi-aspect HRRP datasets from multiple targets. According to the scattering center model, HRRPs from complex targets yield target signatures that are a strong function of the target-sensor orientation [3]–[7], [10]–[14], referred to as target-aspect sensitivity in [4]–[7], [12]. To deal with the target-aspect sensitivity, the literature [5], [6], [12], [13] divided the multi-aspect HRRP dataset from each target into subsets, of which the HRRP data in each subset are collected from a target-aspect sector roughly without the scatterers' motion through range cells (MTRC). Thus each subset is defined to be an aspect-frame from the



(a) (b) (c)

Fig. 1. Real HRRP samples and corresponding FFT-magnitude feature vectors of different airplane targets at the target-aspect angle of 165^o. (a) Yark-42; (b) Cessna Citation S/II; (c) An-26.

target. To model the FFT-magnitude feature vectors of a multi-aspect HRRP dataset from target c ($c \in \{1, 2, \cdots, C\}$ with C denoting the number of targets) here, we accordingly divide the feature dataset into M_c aspect-frames, *e.g.,* the mth frame (here $m \in \{1, \cdots, M_c\}$) is $\{\boldsymbol{y}^{(c,m)}(n)\}_{n=1}^{N}$ with N denoting the number of samples, $\boldsymbol{y}^{(c,m)}(n) = [|\mathcal{F}_1^{(c,m)}(n)|, |\mathcal{F}_2^{(c,m)}(n)|, \cdots, |\mathcal{F}_L^{(c,m)}(n)|]^{\mathrm{T}}$ representing the nth FFT-magnitude feature vector in this frame, and L denoting the number of frequency bins in a FFT-magnitude feature vector. If the learning of FA model for the mth aspect-frame from target c, *i.e.*, $\{\boldsymbol{y}^{(c,m)}(n)\}_{n=1}^{N}$, is termed the single task FA (STL-FA) modeling for the frame m from target c, here we wish to construct a MTL-FA model for all M_c frames from target c jointly.

Our hierarchical model for the cth target with $c = 1, 2, \cdots, C$ may be represented as

$$
\begin{aligned}
&\boldsymbol{y}^{(c,m)}(n) = \boldsymbol{A}^{(c)}\boldsymbol{w}_n^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\epsilon}_n^{(c,m)} \\
&\boldsymbol{w}_n^{(c,m)} = \widehat{\boldsymbol{w}}_n^{(c,m)} \circ \boldsymbol{z}^{(c,m)} \\
&\boldsymbol{z}^{(c,m)} = [z_1^{(c,m)}, z_2^{(c,m)}, \cdots, z_K^{(c,m)}]^{\mathrm{T}} \\
&\widehat{\boldsymbol{w}}_n^{(c,m)} \sim \mathrm{Norm}(0, \gamma^{-1}\boldsymbol{I}_K)\ , \quad \gamma \sim \mathrm{Ga}(d_0, e_0) \\
&z_k^{(c,m)} \sim \mathrm{Ber}(\pi_k^{(c,m)}) \\
&\pi_k^{(c,m)} \sim \mathrm{Beta}(a_0/K, b_0(K-1)/K) \\
&\boldsymbol{A}_{\cdot k}^{(c)} \sim \mathrm{Norm}(\boldsymbol{0}, \frac{1}{L}\boldsymbol{I}_L)\ , \quad \boldsymbol{\mu}^{(c,m)} \sim \mathrm{Norm}(\boldsymbol{\mu}_0^{(c,m)}, \frac{1}{N}\boldsymbol{I}_L) \\
&\boldsymbol{\epsilon}_n^{(c,m)} \sim \mathrm{Norm}(\boldsymbol{0}, \mathrm{diag}(1/\alpha_1^{(c,m)}, 1/\alpha_2^{(c,m)}, \cdots, 1/\alpha_L^{(c,m)})) \\
&\alpha_f^{(c,m)} \sim \mathrm{Ga}(g_0, h_0) \\
&\qquad n = 1, 2, \cdots, N\ ;\ m = 1, 2, \cdots, M_c \\
&\qquad k = 1, 2, \cdots, K\ ;\ f = 1, 2, \cdots, L
\end{aligned}
\tag{3}
$$

where $\boldsymbol{y}^{(c,m)}(n) \in \mathbb{R}^L$ denotes the nth L-dimensional FFT-magnitude feature vector in the mth frame of the cth target; $\boldsymbol{A}^{(c)} = [\boldsymbol{A}_{\cdot 1}^{(c)}, \boldsymbol{A}_{\cdot 2}^{(c)}, \cdots, \boldsymbol{A}_{\cdot K}^{(c)}] \in \mathbb{R}^{L \times K}$ is the shared factor loading matrix of the cth target with $\boldsymbol{A}_{\cdot k}^{(c)}$ for $k = 1, \cdots, K$ denoting the kth column of $\boldsymbol{A}^{(c)}$ and K being an integer chosen to be large relative to the number of anticipated factors (*e.g.*, we may set $K = 2 \times L$ in the following experiments); $\boldsymbol{w}_n^{(c,m)} = \widehat{\boldsymbol{w}}_n^{(c,m)} \circ \boldsymbol{z}^{(c,m)} \in \mathbb{R}^K$ is the latent factor variable of the nth feature vector in the mth frame of the cth target, with $\circ$ representing the point-wise (Hadamard) vector product and $\boldsymbol{z}^{(c,m)} = \{z_k^{(c,m)}\}_{k=1}^{K}$ being a binary vector for selecting the columns of $\boldsymbol{A}^{(c)}$ for the mth frame; $\boldsymbol{\mu}^{(c,m)} \in \mathbb{R}^L$ is the mean variable of the mth frame of the cth target; $\boldsymbol{\epsilon}_n^{(c,m)} \in \mathbb{R}^L$ is the noise variable of the nth feature vector in the mth frame of the cth target, with the diagonal components of the diagonal precision matrix (inverse of variance) $\boldsymbol{\alpha}^{(c,m)} = \{\alpha_f^{(c,m)}\}_{f=1}^{L} \in \mathbb{R}^+$; $\mathrm{diag}(u_1, u_2, \cdots, u_L)$ denotes a diagonal matrix with the vector $\boldsymbol{u} = \{u_f\}_{f=1}^{L}$ on the diagonal; $\boldsymbol{I}_K$ is the $K \times K$ dimensional identity matrix (with $\boldsymbol{I}_L$ similarly defined); a_0, b_0, d_0, e_0, g_0, h_0 and $\{\boldsymbol{\mu}_0^{(c,m)}\}_{m=1}^{M_c}$ are preset hyper-parameters. We use $\mathrm{Norm}(\cdot)$, $\mathrm{Ga}(\cdot)$, $\mathrm{Ber}(\cdot)$ and $\mathrm{Beta}(\cdot)$ to refer respectively to Normal, Gamma, Bernoulli and Beta densities parameterized by the arguments inside parentheses.

In the MTL-FA model for the FFT-magnitude feature of HRRP data, the factor loading matrix is assumed to be shared across all aspect-frames from a given target, but the latent factor variables are target-aspect dependent. This implies that the columns of the factor loading matrix commonly selected by different target-aspect frames are learned jointly via the feature data of these frames. By contrast with learning a unique factor loading matrix for each frame in [4], such MTL mechanism offers the potential to more accurately estimate the factor loading matrix via more related data. In addition, it is important to emphasize that it is the time-shift invariant feature, *i.e.*, the FFT-magnitude feature, makes it possible to share the same factor loading matrix across all the target-aspect frames from each target; otherwise, the problem of the time-shift compensation between different target-aspect frames needs to be dealt with before the statistical modeling. Since the scatterers' MTRC occurs between different target-aspect frames, the conventional time-shift compensation method cannot be used here and in theory it is very difficult to compensate the time-shift between aspect-frames.

In the training phase, we seek to estimate the posterior distribution of the latent variables $\boldsymbol{\Psi}$ in the model, given the observed training data $\boldsymbol{Y}$ and the hyper-parameters $\boldsymbol{\Upsilon}$, via Bayesian inference. We employ variational Bayesian (VB) [1] inference as a compromise between accuracy and efficiency. This method approximates an intractable joint posterior $p(\boldsymbol{\Psi}|\boldsymbol{Y})$ of all the hidden variables by a product of marginal distributions $q(\boldsymbol{\Psi}) = \prod_h q_h(\Psi_h)$, each over only a single hidden variable Ψ_h. The optimal parameterization of $q_h(\Psi_h)$ for each variable is obtained by minimizing the Kullback-Leibler divergence between the variational approximation $q(\boldsymbol{\Psi})$ and the true joint posterior $p(\boldsymbol{\Psi})$.

C. Noise-robust classification

After learning the model for each target c with $c = 1, 2, \cdots, C$, $q(\boldsymbol{\Psi}^{(c)})$ is obtained. If a test HRRP sample $\boldsymbol{x}^*$ is measured under the same SNR circumstance as that of training data, the frame-conditional likelihood of the corresponding FFT-magnitude feature $\boldsymbol{y}^*$ given the frame m of the target c is

$$p(\boldsymbol{y}^*|m, c) = \text{Norm}(\boldsymbol{\chi}^{(c,m)}, \boldsymbol{\Omega}^{(c,m)}) \tag{4}$$

with

$$\left\{\begin{array}{l} \boldsymbol{\chi}^{(c,m)} =< \boldsymbol{\mu}^{c,m} > + \\ < \boldsymbol{A}^{(c)} > \text{diag}(< \boldsymbol{z}^{(c,m)} >)(\frac{1}{N}\sum_{n=1}^{N} < \widehat{\boldsymbol{w}}_n^{(c,m)} >) \\ \boldsymbol{\Omega}^{(c,m)} =< \boldsymbol{A}^{(c)}\text{diag}(\boldsymbol{z}^{(c,m)}) \\ \{\frac{1}{N}\sum_{n=1}^{N}[(\widehat{\boldsymbol{w}}_n^{(c,m)} - < \widehat{\boldsymbol{w}}_n^{(c,m)} >) \\ (\widehat{\boldsymbol{w}}_n^{(c,m)} - < \widehat{\boldsymbol{w}}_n^{(c,m)} >)^{\mathrm{T}}]\}\text{diag}(\boldsymbol{z}^{(c,m)})\boldsymbol{A}^{(c)\mathrm{T}} > \\ +\text{diag}(< \boldsymbol{\alpha}^{(c,m)} >)^{-1} \end{array}\right. \tag{5}$$

where $< \cdot >$ means the posterior expectation for the latent variable over the corresponding distribution on it.

In the above discussion, we assume the noise level of the test sample is same as those of the training samples, the statistical recognition method is referred to as *MTL-FA model without modification in classification* in this paper. However, for RATR problem, the training data can usually be obtained under high SNR via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to the large distance between the non-cooperative targets and radar. If we assume that the measured test sample under low SNR can be expressed as $\widetilde{\boldsymbol{y}}^* = \boldsymbol{y}^* + \boldsymbol{\nu}^*$, where the noise level of $\boldsymbol{y}^*$ is same as those of the training samples (namely with high SNR) and $\boldsymbol{\nu}^*$ is the new noise variable introduced from the test condition with $\boldsymbol{\nu}^* \sim \text{Norm}(\boldsymbol{0}, r_*^{-1}\boldsymbol{I}_L)$, we can have

$$p(\widetilde{\boldsymbol{y}}^*|m, c) = \text{Norm}(\boldsymbol{\chi}^{(c,m)}, \widetilde{\boldsymbol{\Omega}}^{(c,m)}) \tag{6}$$

with

$$\begin{array}{l} \widetilde{\boldsymbol{\Omega}}^{(c,m)} =< \boldsymbol{A}^{(c)}\text{diag}(\boldsymbol{z}^{(c,m)}) \\ \{\frac{1}{N}\sum_{n=1}^{N}[(\widehat{\boldsymbol{w}}_n^{(c,m)} - < \widehat{\boldsymbol{w}}_n^{(c,m)} >)(\widehat{\boldsymbol{w}}_n^{(c,m)} - < \widehat{\boldsymbol{w}}_n^{(c,m)} >)^{\mathrm{T}}]\} \\ \text{diag}(\boldsymbol{z}^{(c,m)})\boldsymbol{A}^{(c)\mathrm{T}} > +\text{diag}(< \boldsymbol{\alpha}^{(c,m)} >)^{-1} + r_*^{-1}\boldsymbol{I}_L \end{array} \tag{7}$$

Since the above noise-robust classification method utilizes the test noise level information r_*, the statistical recognition method is referred to as *noise-robust MTL-FA model with test noise level prior* in this paper. Although the MTL-FA model is simply modified to match the new test circumstance in the classification stage via the above approach, it is hard to accurately estimate the noise level prior in the real application. As shown in the following experiments, the recognition performance degrades dramatically, when the prior has just $\pm$5dB bias.

Another approach of the noise-robust classification based on MTL-FA model is to adaptively learn a new posterior distribution of the noise precision variable in the MTL-FA model given the noised test sample $\widetilde{\boldsymbol{y}}^*$ for each frame of each target, *i.e.*, $q(\boldsymbol{\alpha}_*^{(c,m)})$ with $m = 1, 2, \cdots, M_c$ and $c = 1, 2, \cdots, C$. We can assume $1/ < \alpha_{*f}^{(c,m)} >= 1/r_* + 1/ < \alpha_f^{(c,m)} >$ with $f = 1, 2, \cdots, L$. Nevertheless, since here we directly learn the $< \boldsymbol{\alpha}_*^{(c,m)} >$, the r_* may be different for different frequency bin f. Therefore, in this approach we directly use $\text{diag}(< \boldsymbol{\alpha}_*^{(c,m)} >)^{-1}$ instead of the last two terms of $\widetilde{\boldsymbol{\Omega}}^{(c,m)}$ in (7), *i.e.*, $\text{diag}(< \boldsymbol{\alpha}^{(c,m)} >)^{-1} + r_*^{-1}\boldsymbol{I}_L$, to calculate the corresponding $\widetilde{\boldsymbol{\Omega}}^{(c,m)}$. This statistical recognition method is referred to as *noise-robust MTL-FA model with the adaptively learned test noise level* in this paper.

III. Experimental Results

We examine the performance of the proposed model on the measured airplane data used in [4], [6]. The center frequency and bandwidth of the radar are 5520MHz and 400MHz. The projections of target trajectories onto the ground plane are shown in Figure 2. In order to test the generalization performance of the recognition methods, the 2nd and the 5th segments of Yak-42, the 6th and the 7th segments of Cessna, the 5th and the 6th segments of An-26 are taken as the training samples, other data segments are taken as test samples in our experiments. Same as those in our previous work [4]–[7], we use the aspect-frame partition method to deal with the target-aspect sensitivity of the training dataset. There are 35 aspect-frames for Yark-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each aspect-frame totally has 1024 HRRP samples, *i.e.*, $C = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, $N = 1024$. In the following experiments, we will compare the recognition performance via training the proposed statistical model with different training data sizes, *i.e.*, different numbers of training samples from each frame (different N).

This HRRP dataset is measured under high SNR. As discussed in Sections I and II-C, for RATR problem, we usually assume that the training data are under high SNR and the test samples are got under the different condition of low

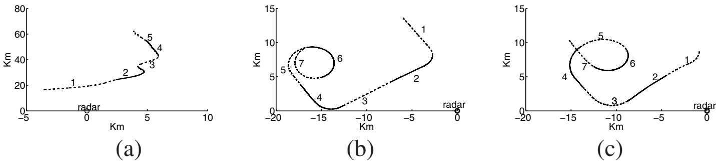


Fig. 2. Projections of target trajectories onto the ground plane: (a) Yark-42; (b) Cessna Citation S/II; (c) An-26.

SNR. For aircraft-like targets under a look-up scenario such as the case of this dataset, the noises in the inphase and quadrature echoes of targets can be assumed to be Gaussian white noises. Therefore, in order to evaluate the robustness of our proposed method, we add simulated Gaussian white noises to the inphase and quadrature components of the raw complex HRRP test data. The SNR is defined as

$$\mathrm{SNR} = 10 \times \log_{10}\left(\frac{\bar{P}_{\boldsymbol{x}}}{P_{\mathrm{Noise}}}\right) = 10 \times \log_{10}\left(\frac{\sum_{l=1}^{L} P_{x_l}}{L \times P_{\mathrm{Noise}}}\right) \tag{8}$$

where $\bar{P}_{\boldsymbol{x}}$ denotes the average power of original HRRP, P_{x_l} denotes the power of the original echo from the lth range cell, L denotes the number of range cells (here $L = 256$), and P_{Noise} denotes the power of noise. Thus $10^{\frac{\mathrm{SNR}}{10}}$ equals to the rate of the average power of original HRRP to the power of the noise.

The recognition accuracies of the MTL-FA model without modification in classification, the noise-robust MTL-FA model with the matched test noise level prior, the noise-robust MTL-FA model with the mismatched test noise level prior and the noise-robust MTL-FA model with the adaptively learned test noise level, versus the size of training dataset for the test data under the condition of SNR$= 15\mathrm{dB}$ are shown in Figure 3, of which the 2nd and 3rd methods utilize the prior information about test noise level (one is accurately $15\mathrm{dB}$ and the other is $10\mathrm{dB}$). It is important to restate that all these four methods are trained with the training data under high SNR, but the classification methods are different for the test data under the condition of SNR$= 15\mathrm{dB}$ as discussed in Section II-C. Here we utilized the five training datasets: 64×135, 128×135, 256×135, 512×135, 1024×135, where 64×135 means that 64 feature vectors randomly selected from each of the 135 aspect-frames, *i.e.,* there are totally 8640 training samples, similarly for other data size expressions. The recognition rates in Figure 3 are averaged over ten runs, and the error bars denote the corresponding standard deviations. From Figure 3, we can clearly see that the usage of the matched test noise level prior or the adaptively learned test noise level in the classification stage can improve the recognition performance of the MTL-FA model trained under high SNR, but the mismatched test noise prior will degrade the recognition performance. In addition, for our MTL-FA model, when the training data size is larger than 256×135, the recognition performance generally remains stable; while for STL-FA model such training data size of each target-aspect frame ($N = 256$) is not enough for the full covariance estimation for the 256-length HRRP vector via separately learning from each target-aspect frame. Thus the better learning performance with small data size is an advantage of MTL over STL. The reason why the recognition rates obtained via the modification with the matched test noise level prior are lower than those via the adaptive modification under the test condition of SNR$= 15\mathrm{dB}$ will be explained in the following paragraph after showing the variation of the recognition performance with SNR in Figure 4.

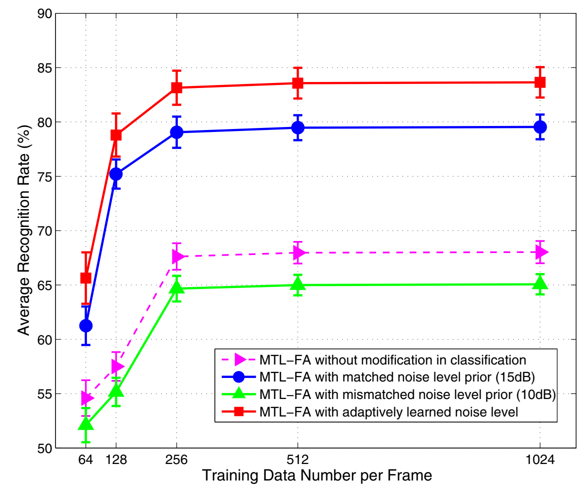


Fig. 3. Variation of the recognition performance with training data size, via the MTL-FA model without modification in classification, the noise-robust MTL-FA model with the matched test noise level prior ($15\mathrm{dB}$), the noise-robust MTL-FA model with the mismatched test noise level prior ($10\mathrm{dB}$) and the noise-robust MTL-FA model with the adaptively learned test noise level, for the test data under the condition of SNR$= 15\mathrm{dB}$.

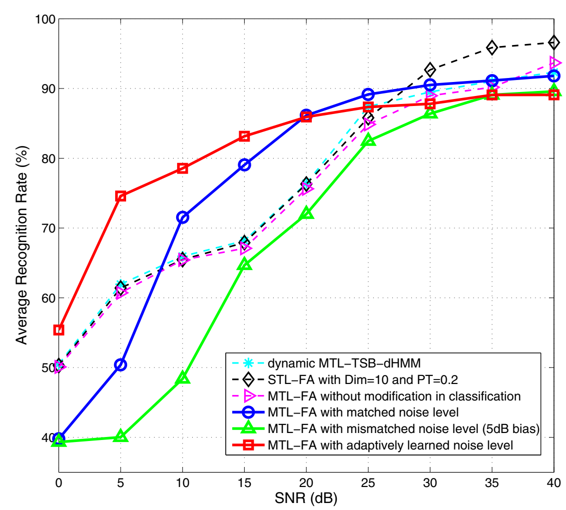


Fig. 4. Variation of the recognition performance with SNR, via the dynamic MTL-TSB-HMM [8], the STL-FA model with the dimensionality of the latent factor variable (*i.e.*, number of factors) $J = 10$ and power transformation $p = 0.2$ [4], the MTL-FA model without modification in classification, the noise-robust MTL-FA model with the matched test noise level prior, the noise-robust MTL-FA model with the mismatched test noise level prior ($\pm 5\mathrm{dB}$ bias), and the noise-robust MTL-FA model with the adaptively learned test noise level.

Figure 4 depicts the average recognition rates versus SNR, generated via the six statistical recognition methods, the dynamic MTL-TSB-HMM model [8], the STL-FA model [4], the MTL-FA model without modification in classification and three versions of the noise-robust MTL-FA model, *i.e.*, the noise-robust MTL-FA model with the matched test noise level prior, the noise-robust MTL-FA model with the mismatched test noise level prior ($\pm 5\mathrm{dB}$ bias) and the noise-robust MTL-FA model with the adaptively learned test noise level. All these methods are trained with the training data under high SNR, but the classifications are implemented under different SNR conditions. Here the first two models are directly used for the real HRRP samples and the four MTL-FA models are for the FFT-magnitude feature extracted from the complex HRRP data. For brevity in Figure 4, we omit the recognition curves from the three traditional statistical models for the real HRRP data, *i.e.*, Gaussian model [11], Gamma model [3] and the two-distribution compounded model [6], which can be found in our papers [6], [8]. From [8], the STL-FA model and the dynamic MTL-TSB-HMM model obviously outperform the three traditional models, especially under the condition of low SNR. When SNR$\geqslant 30\mathrm{dB}$, the STL-FA model with power transformation outperforms other five models; when SNR$\geqslant 25\mathrm{dB}$, our noise-robust MTL-FA model with the matched test noise level prior is a little superior to that of the noise-robust MTL-FA model with the adaptively learned test noise level; when SNR is $25\mathrm{dB}$, the noise-robust MTL-FA model with the matched test noise level prior is better than other methods; when SNR is $20\mathrm{dB}$, the noise-robust MTL-FA models with the matched test noise level prior and the adaptively learned test noise level obtain the similar recognition rates, which are much higher than those obtained via the other methods; when SNR$\leqslant 15\mathrm{dB}$, the noise-robust MTL-FA model with the adaptively learned test noise level yields the best recognition rates. The reason why the matched noise level prior cannot yield the best performance when SNR$\leqslant 15\mathrm{dB}$, which can also be observed in Figure 3 for SNR$= 15\mathrm{dB}$, is that the noise components in HRRP data are complex Gaussian distributed, but the FA model assumes the noise a real Gaussian variable. Although we still use the FA structure in the adaptive method, the re-learned posterior distribution of the noise precision variable can adjust the statistical model to match the noised test data better than that directly modified by adding a real noise component as in the method using the test noise level prior. If we assume that the average recognition rate threshold for real application is 70%, our noise-robust MTL-FA model with the adaptively learned test noise level can work under the test condition of SNR$\geqslant 5\mathrm{dB}$; while the STL-FA model and the dynamic MTL-TSB-HMM model need SNR$\geqslant 20\mathrm{dB}$. Thus our noise-robust method can significantly increase the recognition distance between targets and radar in the real application. In this recognition experiments, our MTL-FA model and STL-FA model are trained with $1024 \times 135 = 34560$ training data; while the dynamic MTL model with $256 \times 135 = 138240$ training data. Here it is important to point out that our MTL-FA model with smaller training data size, *e.g.*, 512×135 or 256×135, can obtain the similar recognition performance as shown in Figure 4, which has been demonstrated via the experiment shown in Figure 3; while the STL-FA model must require the training data more than 512×135, due to the full covariance estimation for the 256-length HRRP vector via separately learning from each target-aspect frame.

IV. Conclusion

We have developed a noise-robust factor analysis model based on multitask learning (noise-robust MTL-FA) for the FFT-magnitude feature extracted from complex HRRP data. The initial-phase and time-shift invariant feature allows the factor loading matrix shared for all target-aspects of each target via the MTL construction (here the multiple tasks are linked to specific aspect-frames of HRRPs from each target). Moreover, the proposed framework can update the noise level parameter in the FA model in the classification phase to adaptively match that of the received test sample. The RATR experiments based on measured HRRP data show that the proposed methods have inspiring recognition performance with small training data size and under the test condition of low SNR.

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